POTATO PLANT LEAF DISEASE DETECTION USING CUSTOM CNN DEEP NET: A STEP TOWARDS SUSTAINABLE AGRICULTURE

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ABSTRACT

Detecting potato plant leaf diseases using convolutional neural networks (CNNs) is a significant step towards sustainable agriculture. Computer vision-based automated disease detection in agriculture is essential for the early detection and treatment of plant diseases, assisting farmers in minimizing crop losses, maximizing yields, and guaranteeing food security. This study proposes a novel method for detecting potato leaf disease using a customized 5-layer Convolutional Neural Network (CNN). The model's performance is compared with MobileNet and a 4-layer CNN as the backbone architectures. Based on experimental results, the 5-layer customized CNN achieves an accuracy rate of 97.16%, which is significantly higher than that of the 4-layer CNN (73.21%) and MobileNet (78.43%). Additionally, the 5-layer CNN model shows promising results for other evaluation metrics, including F1 score (97.18%), recall (97.16%), and precision (97.24%). Furthermore, out of all the models that were tested, the 5-layer CNN model shows the least amount of loss. Using a threshold of 0.6, and the custom CNN as backbone architecture to Faster R-CNN (FRCNN) the model achieved an Intersection over Union (IoU) of 0.76 for disease detection. Additionally, a comparative study of various optimizers (Adam, SGD, Adadelta, and AdamW) and loss functions is done; the Adam optimizer and the unique 5-layer CNN model yielded the best results. This study advances automated methods for detecting potato leaf diseases, offering a dependable and effective way to identify diseases early in agricultural settings.

Keywords – Computer Vision, Convolution Neural Network (CNN), Sustainable agriculture, Early Blight, Late Blight

1. INTRODUCTION

Conference Automation with computer vision enabled with deep learning in agriculture revolutionizes efficiency and productivity from planting to harvesting, ensuring sustainable food production for the future. Convolutional neural networks (CNNs) to detect leaf diseases in potato plants is a big step towards sustainable agriculture. Farmers can reduce the need for widespread pesticide application by implementing targeted treatments in response to early disease identification. By doing this, chemical use is reduced while simultaneously protecting soil quality and avoiding environmental contamination. Furthermore, early disease detection promotes sustainable farming methods, ensures food security, and prevents yield loss. All things considered, using technology to detect diseases helps make potato farming more ecologically friendly and sustainable. Potato (Solanum tuberosum) is one of the most important food crops worldwide, playing a vital role in global agriculture and food security. Not only is it a staple food for millions of people, but it also serves as a significant source of income for farmers and contributes significantly to the economies of many countries. However, potato plants are susceptible to various diseases, which can have detrimental effects on both yield and quality. Early detection and management of these diseases are crucial for sustainable potato production.

In recent years, there has been a growing interest in using technology, particularly deep learning techniques, for the early detection and management of plant diseases. In this paper, we propose a novel approach for potato plant leaf disease detection using a custom-designed Convolutional Neural Network

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(CNN) Deep Net.

In this present work, three different models were implemented for the classification purpose: a custom designed fine-tuned 5-layer CNN, a 4-layer CNN, and MobileNet. These models were used as the backbone for the detection model. Among the three models, the proposed custom 5-layer CNN exhibited the highest accuracy, achieving nearly 97%. Additionally, a comparative analysis of the accuracies was conducted with different optimizers, including Adam, AdamW, and SGD.

For the detection [1] purpose of potato leaf diseases Faster R-CNN, a state-of-the-art object detection algorithm, was implemented. Three categories of potato leaves were considered: Early blight, Late blight, and healthy leaves. These categories were extracted from the Plant Village dataset, a comprehensive database of plant images for research and development.

The best results were yielded by the detection model when using the 5-layer custom CNN model as the backbone, generating minimal model loss. Additionally, the outputs generated after detection can be easily visualized by the user, providing information such as category and score. The best Intersection over Union (IoU) was obtained with the 5- layer CNN as the backbone to the Faster R-CNN, achieving an IoU of 0.76 with a threshold of 0.6. This indicates the robustness of our detection model in accurately identifying and localizing potato leaf diseases.

The organization of the remaining paper is as follows: Section 2 exhibits the literature review, Section 3 highlights the overall methodology, Section 4 show up the result and discussion and finally Section 5 summarize the entire work with the conclusion.

2. RELATED WORKS

An exhaustive study of the existing literature is carried out for the present work and are discussed in this section with the key points. Sofuoglu C I and Birat D in [2] proposed a deep learning model for classifying the plant diseases. The CNN model used, acted as filter to the input images for extraction of the key features and thereafter classification is done. In [3] authors implemented a simple CNN classifier for the classification of the potato leaf diseases with some basic image pre-processing. A pre-trained model with the fine tuning of the hyper-parameters is implemented [4]. The model after training acquired 97.4% of classification accuracy for the categories of early blight and late blight. Islam and Sikdar[5] in their article undergone classification of the diseased and healthy leaves using a conventional CNN classifier. They have endured augmentation process for the increase of the data samples. Anushka et. al [6] proposed the idea of again classification with the CNN model and obtained an accuracy of 91.41%. A comparative analysis for the classification of potato leaf diseases with the existing models viz. Googlenet, Resnet and VGG16 is gone through and obtained 97% model accuracy [7].An effective idea with amalgamation of classical machine learning and deep learning is proposed in [8]. Authors build the model with K-means clustering segmentation followed by VGG-16 for the final classification of the potato leaf disease. An accuracy of 97% is achieved. A 14-layer CNN is utilized for the purpose of classification along with augmentation of the data samples[9]. A 5-class classification on different types of potatoes is carried out in the paper[10] with a CNN model and compared with other models such as Alexnet, Googlenet, VGG, R-CNN. Segmentation with the K-means clustering is used for feature extraction and multi-class support vector machine is implemented for the classification purpose[11]. The accuracy obtained with the kind of classical machine learning technique is 95.99%. All the existing works are undergone majorly with the classification task and some of the works are with the combination of classical machine learning algorithms. In the present work, a novel proposal with a customized CNN model along with Faster Region based CNN(FRCNN) is executed for the detection of diseased potato leaf and the model generated effective results in terms of the performance parameters.

3. METHODOLOGY

The entire workflow of potato leaf disease detection is depicted in Fig. 1. The overall work has been processed using plant Village dataset [12] available at kaggle repository. It comprises images of healthy and diseased leaves of various plants out of which potato plant leaves data were taken. There are 3 categories of leaves-Early blight, Late blight and healthy. The dataset is computed using 3 different architectures viz. Mobile net, 4-layer custom CNN and 5-layer custom fine-tuned CNN. The Accuracy graphs are generated for the three respective architectures. The training is followed by analysis of classification reports for a comparative study amongst the different parameters of the respective architectures. In the next step, detection was carried out where in the classified outputs are fed in to the detection architecture, which in our case is F-RCNN, to get

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desired results for diseased potato leaves.

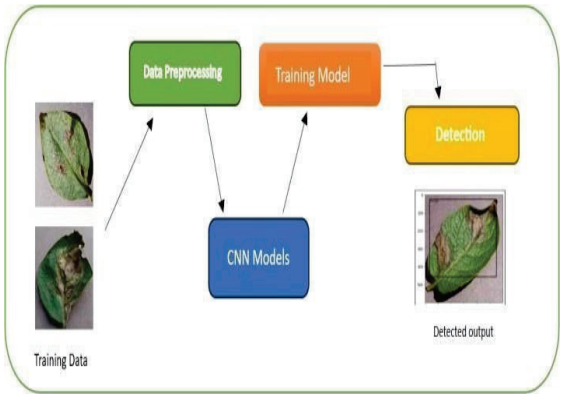

Fig.1 Workflow diagram

3.1 Dataset

The proposed work has been accomplished using plant Village dataset [12] available at Kaggle. Some examples are shown in Fig. 2. The dataset consists of 3 classes viz. late_blight, early_blight and healthy. The dataset contains 1000 images for the diseased classes and 152 images for the healthy leaves. To address the data imbalance problem the healthy category data are augmented and increased to 1000 samples. These 3000 samples were used to train the classifiers. Further, for detection 64 images after augmentation and annotation

Fig.2 Sample Dataset images

were taken to train the FRCNN model. All the images were taken in JPG format for convenience with dimension being [256 x 256 x 3] pixels. The images are of high resolution which helps in clear detection and segregation.

3.2 Data Pre-processing

The dataset considered is made sure to have a balance between each class. In order to do so, healthy plant images were generated from existing dataset by altering its hue and exposure. For the training of the backbone classifier the dataset was split into training and validation sets in the ratio of 70:30 respectively. For detection purposes the images were annotated using roboflow.com. The datasets are annotated and labeled into three classes: late_blight, early_blight and healthy into json format to represent the data (coco dataset format).

3.3.1 4_Layer CNN

An architecture of CNN with four convolutional layers was built in order to serve as the backbone for the Faster R-CNN (FRCNN). Three specific classes were *the focus of the model's training on the Plant Village dataset. Early outcomes showed encouraging possibilities for additional improvement. Categorical cross-entropy was used as the loss function during training, which took place over ten epochs using the Adam optimizer with the learning rate set to 1e-3. To maximize feature extraction and model performance, post-convolutional layers were successively followed by batch normalization and max- pooling techniques. 73.21% was the reported validation accuracy attained. The 4-layer CNN architecture representation is shown in Fig. 3.*

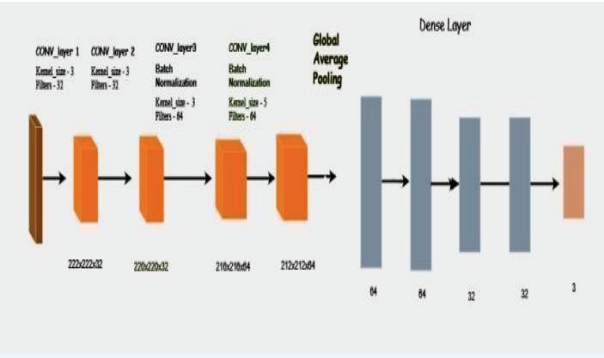

Fig.3 4-layer CNN

3.3.2 5_Layer CNN

The model architecture included five convolutional layers and a convolutional transpose (conv transpose) layer to improve feature extraction and map adjustment. To improve model performance during training, the Adam optimizer was used in conjunction with a categorical cross-entropy loss function. The conv transpose layer was critical in properly constraining and refining the feature maps, allowing for efficient information flow and spatial resolution adjustment throughout the network. The model was trained for 10 epochs with a learning rate of 1e-3. This meticulously tuned configuration produced a remarkable 97% accuracy on the validation dataset. These findings highlight the importance of strategic hyper-parameter selection and the incorporation of advanced architectural components in achieving high performance and accuracy in image analysis tasks. The 5-layer custom CNN architecture representation is shown in Fig. 4.

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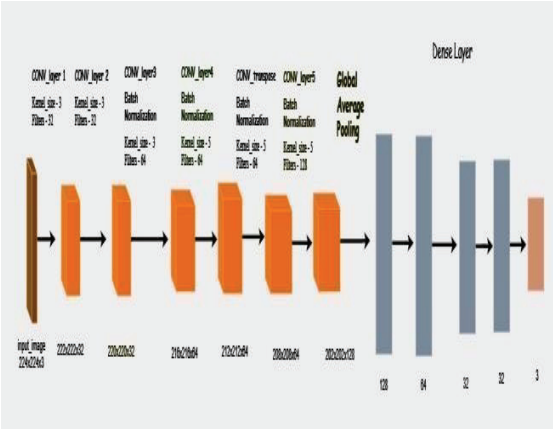

Fig.4 5-layer CNN

3.3.3 Mobile Net

The Mobile Net architecture as shown in Fig. 5 was used to process input images of 224x224x3, using an Adam optimizer with a learning rate of 1e-3 over a training period of 10 epochs. The model was trained with a cross- entropy loss function, yielding an accuracy of 78.43%. Mobile Net is based on depth-wise separable convolutions, a design technique for reducing computational complexity and model size while maintaining performance. This architecture is made up of depth wise convolution followed by point-wise (1x1) convolution layers that efficiently capture spatial and channel-wise dependencies in the input data.

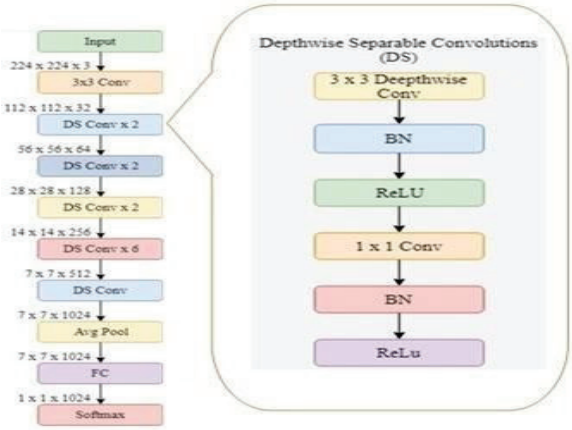

Fig.5 Architecture of Mobile Net

3.3.4 DETECTION

The framework's final stage makes use of Faster R-CNN (F- RCNN), shown in Fig. 6 which has a unique architecture with two networks: a Region Proposal Network (RPN) and an object detection network, to replace traditional selective search methods. F-RCNN consists of three major components. To begin, convolutional layers enable critical feature extraction by employing a variety of filters for microscopic image analysis, which is supplemented by pooling and fully connected layers for accurate classification. The RPN component determines the presence of objects and

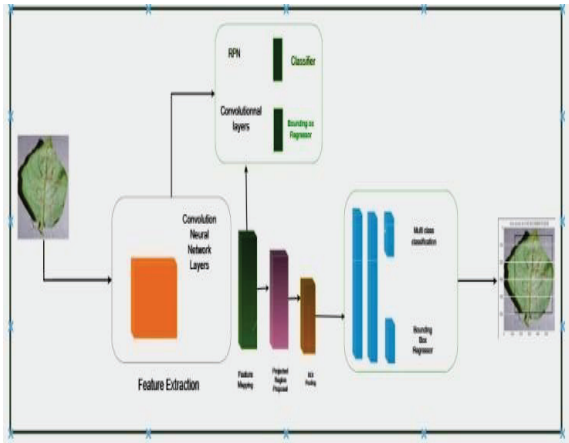

Fig.6 Architecture of FRCNN

predicts precise bounding boxes, while the fully connected neural network processes features to predict object classes. Among the proposed architectures, the 5-layer CNN produced the best results, demonstrating superior accuracy and feature extraction capabilities in the detection framework. This demonstrates the effectiveness of specific design choices in the 5-layer CNN model for comprehensive object detection.

A wide range of pre-trained and specially-designed models were put through extensive testing in this thorough study in order to determine their effectiveness. The 5-layer convolutional neural network (CNN) performed better than the other models on all criterion, showing constant superiority over the others. The major contributions for the 5-layer custom CNN model are listed below:

- A cross-entropy loss function combined with the Adam optimizer, trained across 10 epochs with a precisely calibrated learning rate of 1e-3, were the specific hyperparameters that were used to methodically fine-tune the model.
- One noteworthy aspect of this CNN architecture was the way the channel counts were gradually doubled in each layer until they reached a maximum of 128 channels. This calculated increase in channel capacity made it easier to create feature maps that are expressive and rich, which allowed the model to identify subtle patterns and nuances in the data and eventually led to its exceptional performance in the experimental context.

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4. RESULTS AND DISCUSSION

The model framework, along with a number of performance parameters, has been carefully examined. In images of a mixture of healthy and diseased potato leaves, the current study can reliably identify diseased potato plant leaves. The present work is carried out using a 16GB NVIDIA P100 GPU and Pytorch libraries loaded on the Kaggle platform on kaggle notebook. The Adam optimizer (which yielded the best results in comparison to the other three) was used to train the three distinct architectures which acted as the backbone to the detection model FRCNN. For ten epochs, a batch of size four is employed. The 5-layer CNN validation accuracy was 97.16%, the 4-layer CNN accuracy was 73.21% and the mobilenet validation accuracy was 78.43%. The accuracy comparison graphs are displayed in Fig. 7. The best model (5-layer CNN) then trained over 30 epochs and fed to the FRCNN model shown in Fig. 8. Fig. 9(a) and Fig. 9(b) illustrates the creation of the confusion matrices and precision-recall curves for the optimal architecture. In the present work, the accuracy comparison with the different optimizers viz. Adam, AdamW, Sgd, Adadelta, RMS Prop has also been executed and the best result is obtained with the Adam optimizer as shown in Fig. 10 and reported in table – 1. The performance parameter matrices attained potentially superior values with precision – 97.24%, recall – 97.16% and F1 score – 97.18%. Hence, the results obtained are promising in their values for the proposed 5-layer custom CNN model.

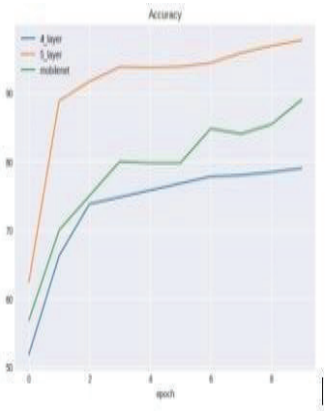


Fig.7 Accuracy Curves for the 3 models

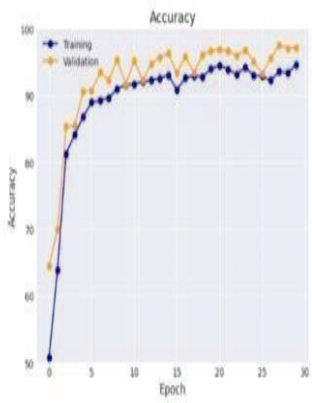
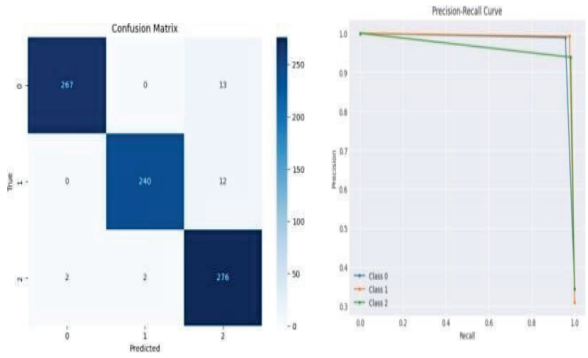


Fig. 8 Training and validation accuracy of 5-Layer CNN over 30 epochs



(a) (b)

Fig. 9 (a) Confusion matrix of 5_layer_cnn (b) Precision-Recall 5_layer_cnn with the 3 classes

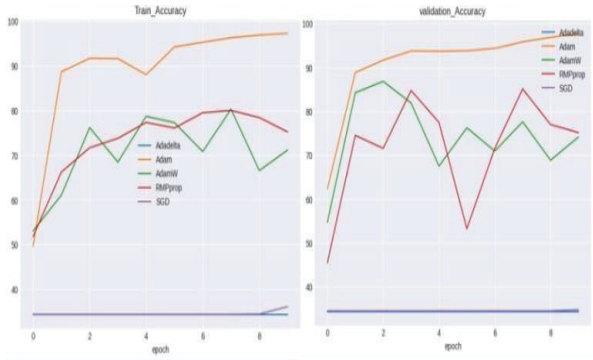


Fig. 10 Training and validation Accuracy comparison curves for 5- layer CNN with different optimizers

Table – 1: Accuracy comparison of 5-layer custom CNN model with different optimizers

Model (with 10 epochs)	Optimizer	Training Accuracy (%)	Validation Accuracy (%)
5-layers CNN	Adadelta	35.76	35.65
	SGD	35.89	38.89
	AdamW	66.946	75.28
	RMS Prop	67.92	78.49
	Adam	**95.89 (97.16 after 30 epochs)**	**96.73 (97.17 after 30 epochs)**

The FRCNN model has been deployed for the detection purpose of the diseased and healthy potato leaves. The model leverages minimum losses with the best model (5-layer custom CNN) in comparison to the other models as reported in table – 2. The best IoU (Intersection over Union) value achieved is 0.78 under a threshold of 0.6. The IoU comparison with the considered classifier models as well as the IoU comparison curves with different optimizers for 5- layer custom CNN has also been plotted and shown in Fig. 11.

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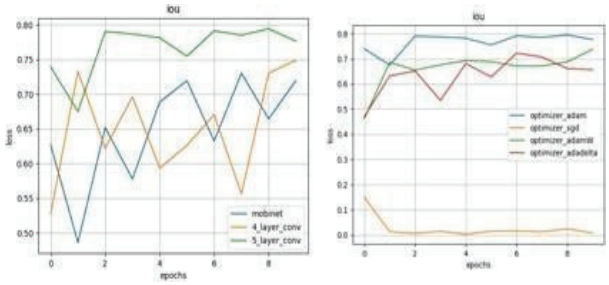
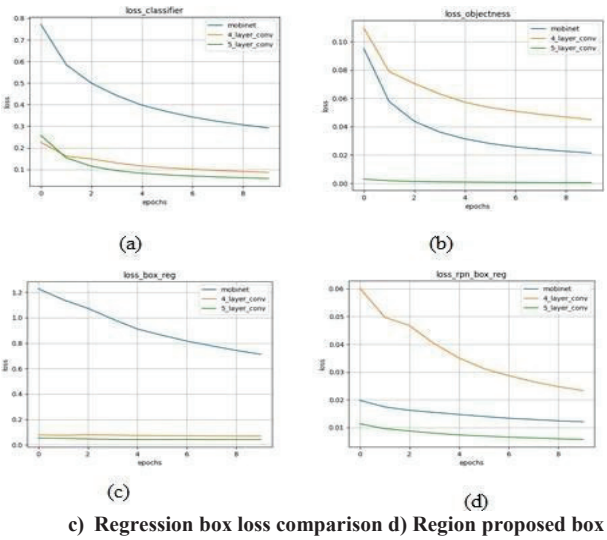


(a) (b)

Fig. 11 (a) IoU curves of the considered classifiers (b) IoU curves with different optimizers for 5-layer custom CNN

A comparison of various losses are done between the considered backbone architectures of FRCNN are shown in Fig. 12(a – d)

Fig. 12(a) Classifier loss comparison b) Objectness loss comparison



(a) (b)

(c) (d)

c) Regression box loss comparison d) Region proposed box loss comparison

Table – 2: Loss and IoU comparison of the 3 considered backbone architectures

Backbone CNN Architecture	Accuracy	Loss_box_reg	Loss_Classifier	Loss_objectness	Loss_rpn_box_reg	IoU
4_layer_cnn	73.21	1.137	0.0698	0.017	0.0089	0.653
5_layer_cnn	**97.16**	**0.0415**	**0.0581**	**0.00052**	**0.0057**	**0.7845**
Mobilenet	78.43	0.924	0.0671	0.013	0.0432	0.648

Some of the detected images of the potato leaves with the 3 classes can be visualized from Fig. 13. The detected images are with the bounding box, the class name and the scores. The current study demonstrates significant utility in the field of agriculture by automating the detection of potato plant leaf diseases. Our custom 5-layer CNN model achieved the best results in terms of accuracy and other performance metrics, as detailed in this section. Furthermore, a comparative analysis with different optimizers revealed promising results for the 5-layer custom CNN, establishing it as the optimal model for this study. Additionally, the FRCNN detection model delivered effective outcomes by minimizing losses associated with various traits. Therefore, we conclude that this novel framework offers a substantial opportunity to enhance the quality of disease analysis and detection in agriculture.

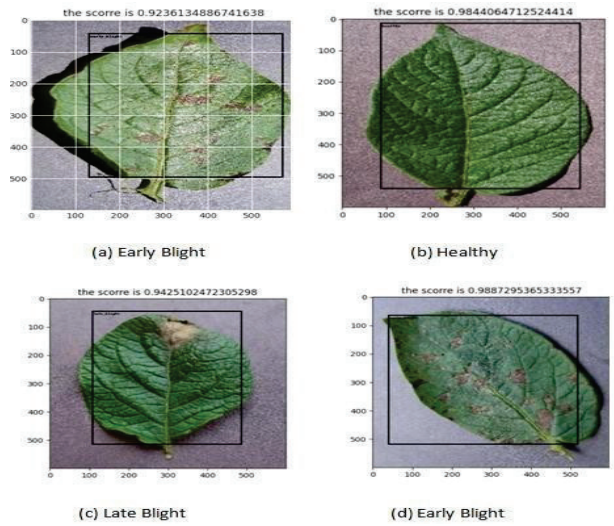


Fig. 13 Sample detected images of diseased and healthy leaves

5. CONCLUSION

Promising results are demonstrated by our proposed approach for potato plant leaf disease detection, which is based on a custom CNN Deep Net. High accuracy in classifying and detecting potato leaf diseases was achieved by leveraging deep learning techniques. For the entire work, three classifiers were considered, namely Mobile Net, a 4-layer CNN, and a 5-layer custom CNN, which served as the backbone architecture to the detection model FRCNN. Among all the models, the best results were generated by the 5-layer CNN, with a classification accuracy of 97.16%, and the best IoU obtained is 0.78 under a threshold of 0.6. An efficient and reliable way to monitor and manage diseases in potato plants is provided by our method, contributing to sustainable agriculture and global food security. In future work, further improvement of the efficiency and accuracy of our detection model is planned by exploring advanced deep learning architectures and optimization techniques. Additionally, the aim is to expand our dataset to include a wider variety of potato diseases, enabling more comprehensive detection and diagnosis capabilities.

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